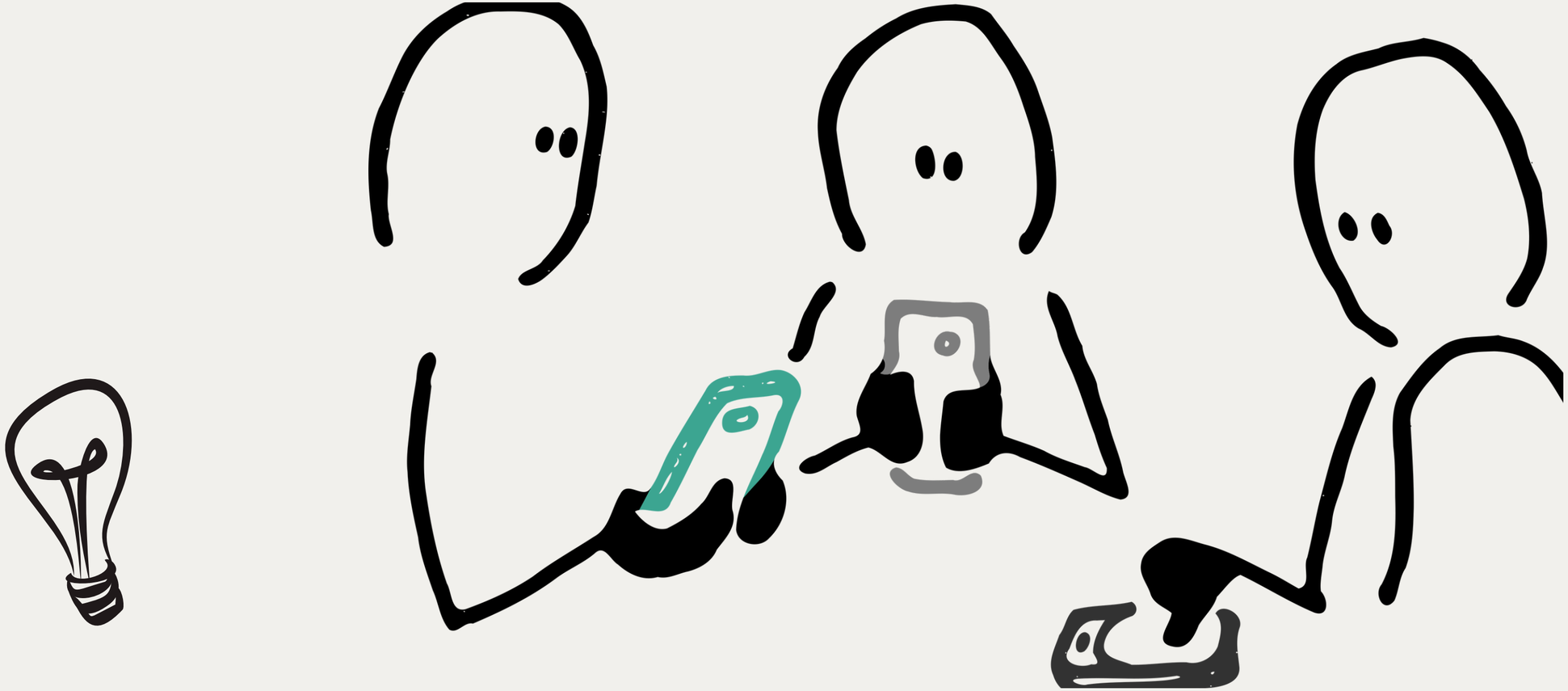


# AI NUTRITION LABEL CHATBOT – SMART FOOD HEALTH ASSISTANT

DEEP LEARNING PROJECT

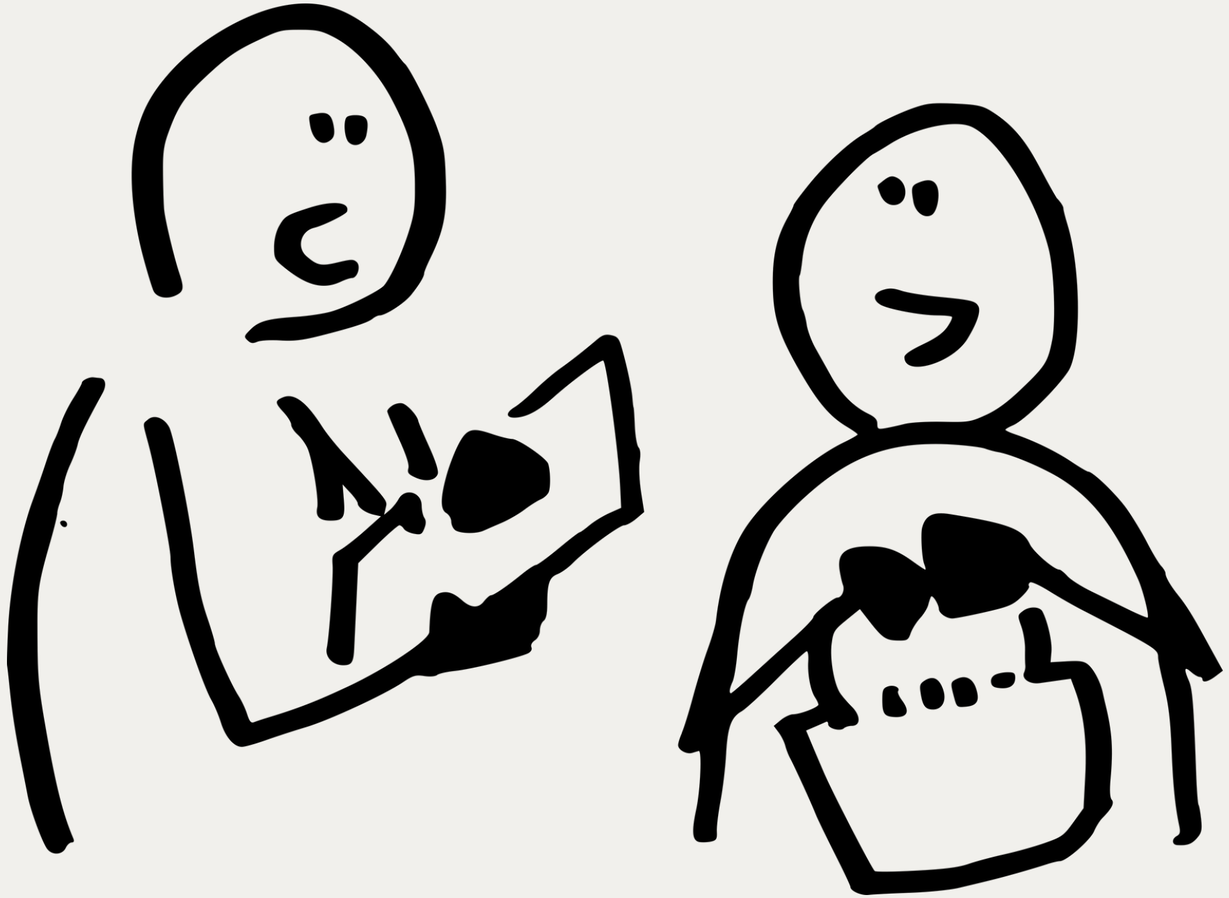
TARAKA NAGA GANESH MUNAGALA.

AKASH VEMULA.

ANAND GOPI SAI KRISHNA

# INTRODUCTION

- NUTRITION LABELS PROVIDE ESSENTIAL INFORMATION ABOUT CALORIES, FATS, SUGARS, SODIUM, AND OTHER NUTRIENTS.
- HOWEVER, MOST CONSUMERS FIND THESE LABELS CONFUSING DUE TO SMALL TEXT, DENSE LAYOUT, AND COMPLICATED TERMINOLOGY.
- WITH INCREASING HEALTH CONSCIOUSNESS, THERE IS A NEED FOR A TOOL THAT CAN INTERPRET LABELS CLEARLY AND INSTANTLY.
- OUR PROJECT AIMS TO BUILD AN AI CHATBOT THAT READS NUTRITION LABELS AND EXPLAINS THEM IN SIMPLE, CONVERSATIONAL LANGUAGE.

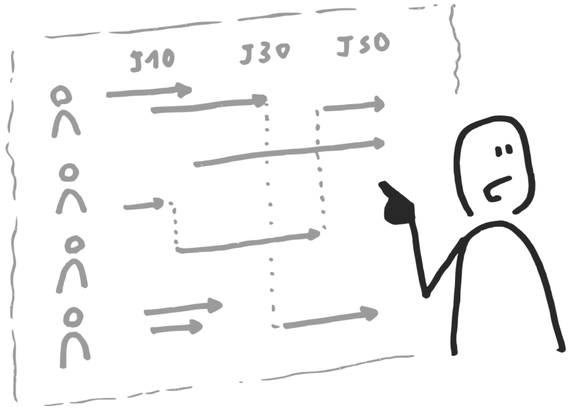


# PROBLEM STATEMENT

- MANY PEOPLE STRUGGLE TO UNDERSTAND WHETHER A PACKAGED FOOD IS HEALTHY OR NOT.
- MANUAL COMPARISON BETWEEN TWO FOOD ITEMS IS TIME-CONSUMING.
- EXISTING APPS ONLY DISPLAY RAW NUTRITION NUMBERS, NOT INTERPRETATIONS.
- NO EXISTING SYSTEM COMBINES:
- ✓ OCR FOR LABEL READING
- ✓ DEEP LEARNING-BASED HEALTH CLASSIFICATION
- ✓ CHATBOT-BASED NATURAL LANGUAGE INTERACTION
- THERE IS A CLEAR NEED FOR A MULTIMODAL SOLUTION FOR BETTER DIETARY DECISIONS.

# MOTIVATION

- LIFESTYLE DISEASES SUCH AS DIABETES, OBESITY, AND HYPERTENSION ARE RISING GLOBALLY.
- 70% OF PACKAGED FOOD USERS RELY ONLY ON GUESSWORK WHILE BUYING SNACKS OR BEVERAGES.
- AI CAN SIMPLIFY NUTRITION BY GIVING PERSONALIZED, REAL-TIME SUGGESTIONS.
- A CONVERSATIONAL SYSTEM MAKES NUTRITION EDUCATION ACCESSIBLE TO EVERYONE.
- THIS PROJECT BRIDGES COMPUTER VISION, NLP, AND HEALTH ANALYTICS TO SUPPORT HEALTHIER EVERYDAY CHOICES.




# OBJECTIVES

**01** Build an OCR pipeline to accurately extract nutrient text from food label images.

**02** Parse and normalize key nutrient values (calories, sugar, sodium, fats, protein, carbs).

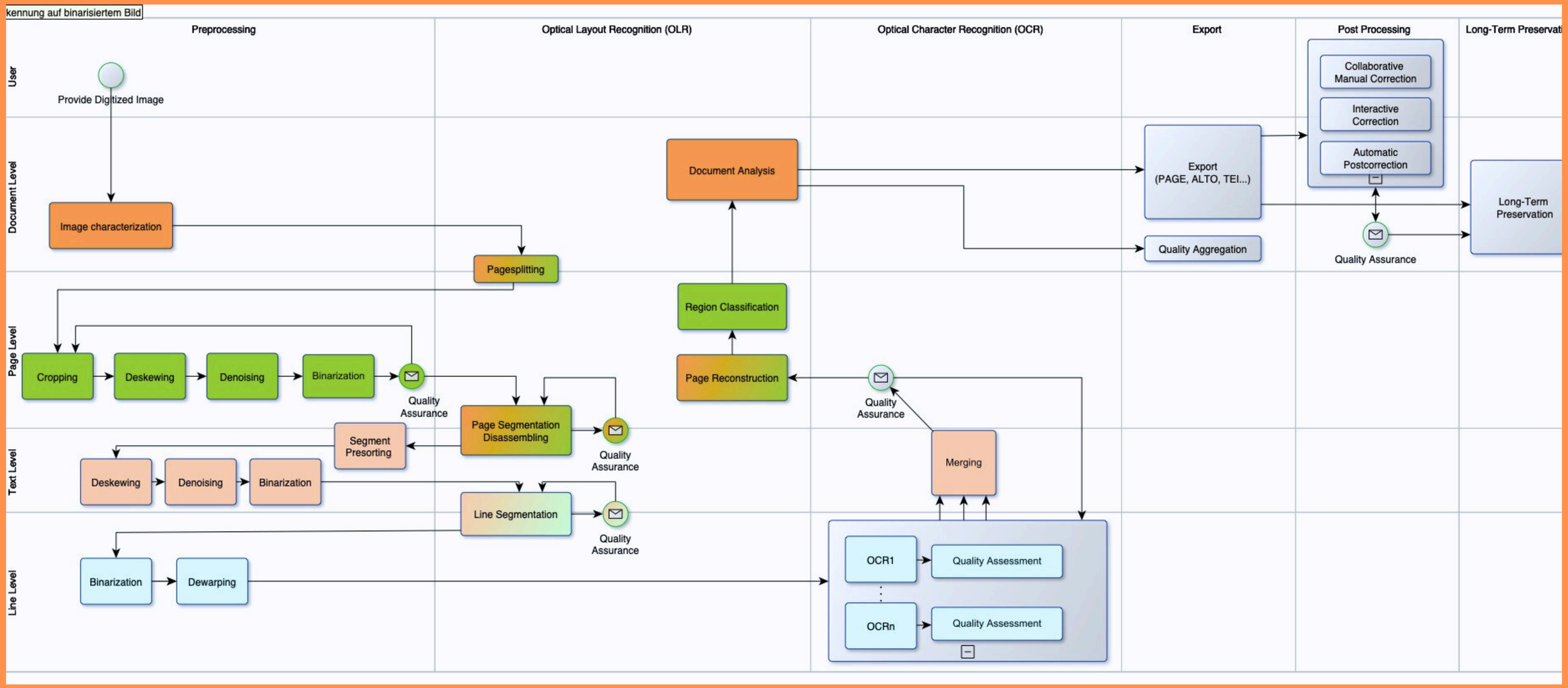
**03** Develop a neural network classifier to categorize food items into Healthy, Moderate, or Unhealthy.

**04** Integrate a chatbot that answers user questions like:

- “Is this item good for daily use?”
- “Is this high in sodium?”
- “Compare this label with my previous snack.”

**05**

- Build a simple, user-friendly interface using Streamlit or Flask.



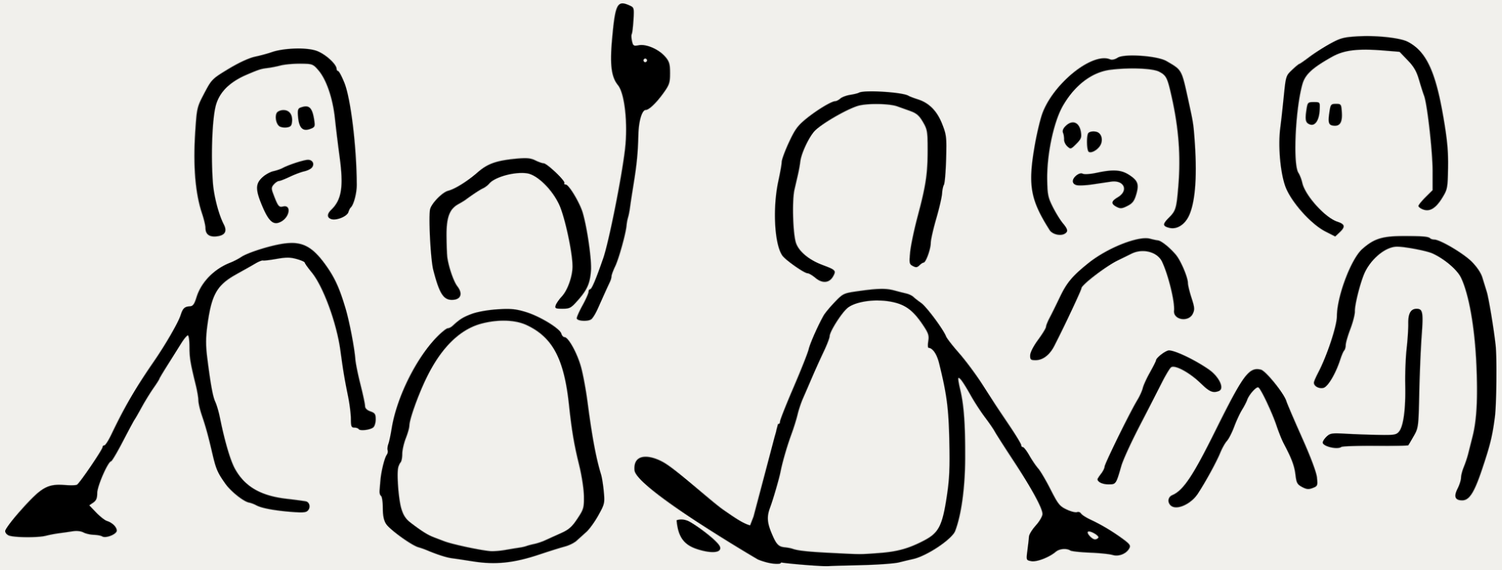

# SYSTEM ARCHITECTURE

- COMBINES COMPUTER VISION (OCR) + MACHINE LEARNING CLASSIFICATION + NLP-BASED REASONING.
- MODULAR DESIGN MAKES IT EASY TO EXTEND FOR MOBILE APPS OR MULTILINGUAL USE.

# DATA COLLECTION

- GOAL : 300-500 REAL NUTRITION LABEL IMAGES.
- SOURCES INCLUDE:
  - KAGGLE FOOD LABEL IMAGE DATASET
  - GROCERY STORE IMAGES (DIFFERENT ANGLES & LIGHTING)
  - PUBLIC DATASETS WITH NUTRITION FACTS
- A SECONDARY TABULAR DATASET CREATED MANUALLY WITH NUTRIENT VALUES EXTRACTED AND LABELED.
- LABELS ARE CATEGORIZED BASED ON WHO/USDA GUIDELINES FOR SUGAR, SODIUM, FATS, AND CALORIES.

# PREPROCESSING



- CONVERT IMAGE TO GRAYSCALE FOR BETTER TEXT CONTRAST.
- REMOVE NOISE USING GAUSSIAN BLUR OR MEDIAN FILTERS.
- ADJUST BRIGHTNESS AND CONTRAST TO ENHANCE OCR ACCURACY.
- APPLY THRESHOLDING (OTSU/ADAPTIVE) TO SEPARATE NUMBERS FROM THE BACKGROUND.
- CORRECT SKEW/TILT USING CONTOUR DETECTION.
- RESIZE ALL IMAGES TO STANDARD DIMENSIONS FOR CONSISTENT OCR RESULTS.

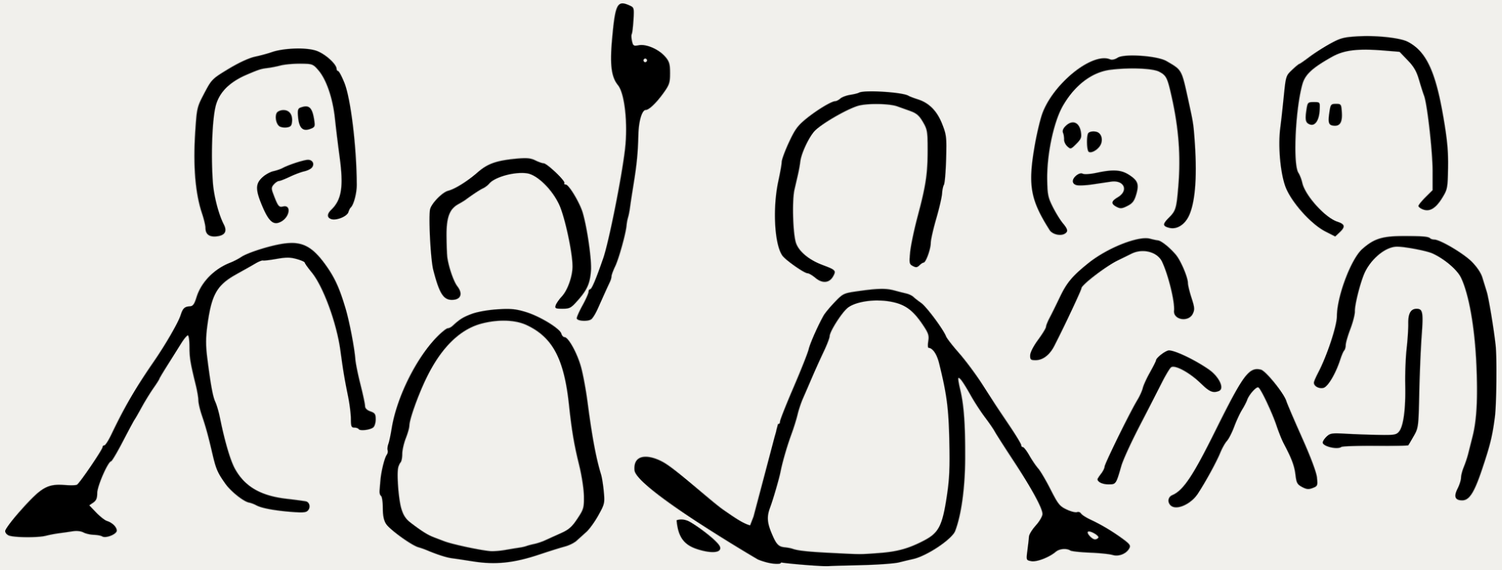
# OCR PIPELINE

- OCR TOOLS USED: TESSERACT OCR AND EASYOCR.
- PIPELINE STEPS:
  - DETECT TEXT REGIONS AND EXTRACT CHARACTER SEQUENCES.
  - USE KEYWORD MATCHING (E.G., “SUGAR”, “FAT”, “CALORIES”).
  - APPLY REGEX TO EXTRACT NUMERIC VALUES AND UNITS.
  - HANDLE UNIT CONVERSION LIKE:
    - MG → G
    - KCAL → CALORIES
  - VALIDATE MISSING VALUES AND FILL USING REASONABLE ASSUMPTIONS.
- OUTPUT: STRUCTURED NUTRIENT DICTIONARY READY FOR ML MODEL INPUT.

# DEEP LEARNING MODEL

- MODEL: MULTI-LAYER PERCEPTRON (ANN)
  - INPUT LAYER: NUTRIENT VALUES
  - HIDDEN LAYERS: DENSE + RELU ACTIVATION + DROPOUT FOR REGULARIZATION
  - OUTPUT LAYER: SOFTMAX (3 CLASSES)
- LOSS FUNCTION: CATEGORICAL CROSS-ENTROPY
- OPTIMIZER: ADAM
- EVALUATION METRICS: ACCURACY, PRECISION, RECALL, F1-SCORE
- MODEL LEARNS TO IDENTIFY PATTERNS SUCH AS:
  - HIGH SODIUM + HIGH SUGAR → UNHEALTHY
  - LOW CALORIES + LOW SUGAR → HEALTHY

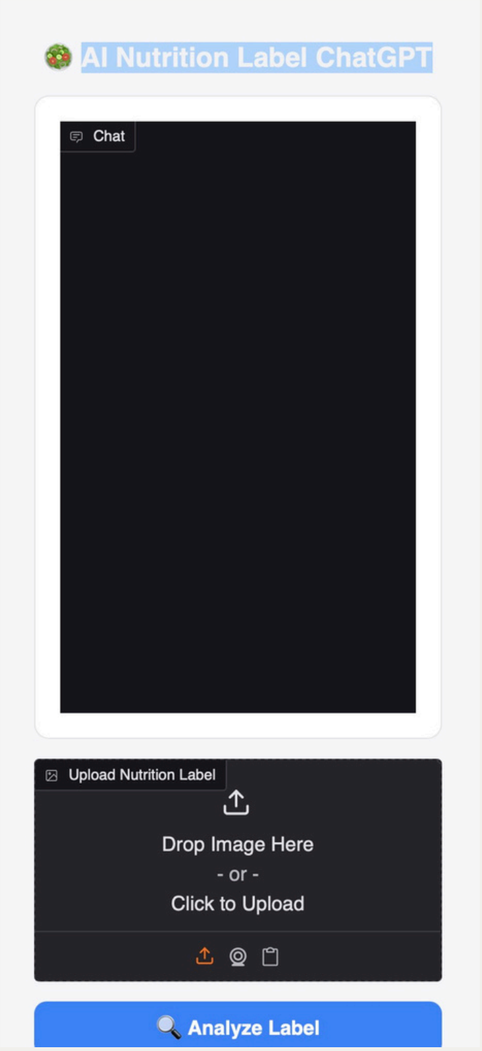
# CHATBOT MODULE



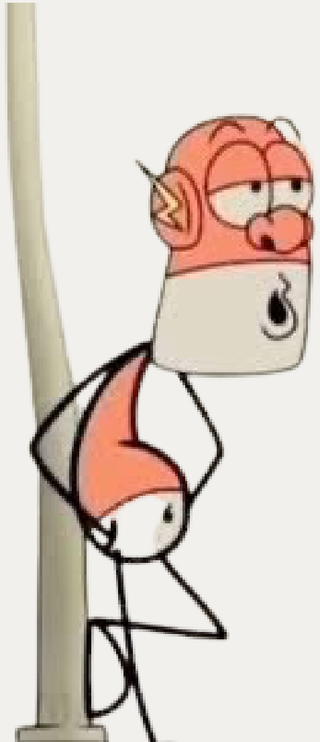
- SUPPORTS CONVERSATIONAL INTERACTION WITH THE USER.
- FEATURES:
- EXPLAINS WHY A PRODUCT IS HEALTHY/MODERATE/UNHEALTHY.
- ANSWERS HEALTH-BASED QUESTIONS LIKE “IS THIS GOOD FOR WEIGHT LOSS?”
- COMPARES TWO OR MORE FOOD LABELS.
- GIVES NUTRIENT-WISE BREAKDOWN (HIGH SUGAR, LOW PROTEIN, ETC.)
- USES RULE-BASED REASONING + KEYWORD INTENT DETECTION.
- DELIVERS EASY-TO-UNDERSTAND HEALTH ADVICE.

# USER INTERFACE (UI)

- BUILT USING STREAMLIT OR FLASK WITH A CLEAN, INTUITIVE LAYOUT.
- KEY COMPONENTS:
  - IMAGE UPLOAD WIDGET
  - CHAT WINDOW FOR CONVERSATION
  - NUTRIENT TABLE
  - HEALTH CATEGORY RESULT
  - EXPLANATION PANEL
- REAL-TIME INTERACTION ENSURES A SMOOTH USER EXPERIENCE.



# CODE DEMO



# RESULTS & EVALUATION

- MODEL PERFORMANCE
  - ACCURACY & F1 SCORE FROM TEST DATASET
  - CONFUSION MATRIX INTERPRETATION

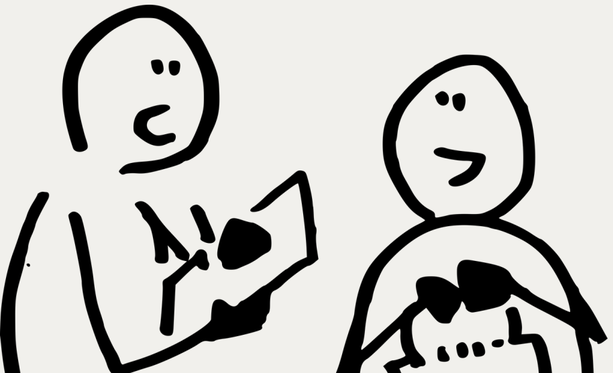
```
...  Test accuracy: 0.9345059394836426
     489/489 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step
     Classification Report:
                   precision    recall  f1-score   support

                0       0.97      0.96      0.96      6789
                1       0.90      0.91      0.90      5240
                2       0.93      0.92      0.92      3606

         accuracy                           0.93     15635
        macro avg       0.93      0.93      0.93     15635
     weighted avg       0.93      0.93      0.93     15635

     Confusion Matrix:
     [[6545  244    0]
      [ 232 4749  259]
      [   0  289 3317]]
```

- OCR PERFORMANCE
  - AVERAGE EXTRACTION ACCURACY ACROSS IMAGES
- SYSTEM PERFORMANCE
  - END-TO-END LATENCY (IMAGE → RESULT)
- EXAMPLE SCREENSHOTS OF CHATBOT INTERACTIONS AND CLASSIFICATION OUTPUT.

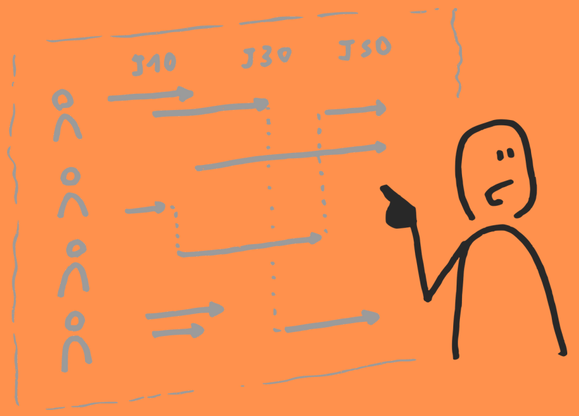


# FUTURE SCOPE

- MULTI-LANGUAGE OCR (SPANISH, FRENCH, HINDI LABELS).
- BARCODE SCANNING TO AUTO-FETCH NUTRITION DATA.
- PERSONALIZED HEALTH PROFILES FOR DIABETES, HYPERTENSION, WEIGHT LOSS.
- RECOMMENDS HEALTHIER ALTERNATIVES BASED ON USER PREFERENCE.
- MOBILE APP EXTENSION USING TENSORFLOW LITE / ONNX RUNTIME.
- INTEGRATION WITH FITNESS/DIET TRACKING APPS.

# CONCLUSION

- DEVELOPED A MULTIMODAL DEEP LEARNING SYSTEM CAPABLE OF INTERPRETING NUTRITION LABELS.
- COMBINES OCR, NUTRIENT ANALYSIS, CLASSIFICATION, AND CONVERSATIONAL AI.
- HELPS USERS MAKE INFORMED DIETARY DECISIONS QUICKLY AND EASILY.
- SERVES AS A FOUNDATION FOR FUTURE HEALTHCARE AND NUTRITION-BASED AI APPLICATIONS.



# THANK YOU!

2025